

First Is Not Stable: Auditing Layerwise Vocabulary Readout in Language Models

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Abstract

Vocabulary projections are often used to ask when an answer becomes visible inside a language model. A common scalar summary—the first layer at which a target enters the vocabulary top- k —has two failure modes: a target that never enters top- k can be conflated with final-layer entry, and a transient entry can be mistaken for stable readout. We audit this measurement and introduce a trajectory-aware protocol that separates first entry, persistent entry, later dropout, and right-censored non-entry. Across 817 TruthfulQA questions and six 7–14B base models, 21.9–31.1% of target tokens never enter the top-10, while 15.2–48.0% enter and later leave it. The scalar first-entry depth hides qualitatively different dynamics: Pythia enters early but is unstable, whereas Gemma enters late and stabilizes almost exclusively at the output. Controls on MMLU and SST-2 further show strong task-format dependence. On SST-2, Mistral and Pythia reach 86.9% and 80.8% candidate-choice accuracy while the correct label appears in the final full-vocabulary top-10 for 0% of examples. We therefore argue that layerwise vocabulary rank is a property of a model-prompt-target interface, not a direct measure of knowledge storage, causal use, or even candidate-task competence. We release corrected code, per-layer ranks, audit checks, and tests in MECHLENS.

1 Introduction

Projecting an intermediate residual state through a language model’s output head gives an intuitive view of how token predictions evolve with depth. This “logit lens” has supported analyses of prediction construction (nostalgebraist, 2020; Geva et al., 2022), factual decoding (Chuang et al., 2024), and early exit (Elhoushi et al., 2024). Its appeal is precisely its simplicity: for a target token, one may record the earliest layer at which it enters the top- k vocabulary predictions.

That scalar is easy to misinterpret. First, targets that never enter top- k are right-censored observations; assigning them the final layer invents an event that did not occur. Second, vocabulary-rank trajectories need not be monotone. A target may enter, leave, and re-enter the top- k , so first entry need not mark a stable transition. Third, the rank is taken over the full vocabulary even when a task evaluates a small candidate set. It can therefore reflect answer formatting and prompt calibration rather than task competence.

We present a measurement audit and a corrected trajectory protocol. We retain the useful question behind the Factual Emergence Point (FEP)—when does a specified target first become directly readable through the model’s native output head?—but explicitly distinguish four outcomes: first entry, persistent entry, dropout after entry, and never observed. We also extract the target by jointly tokenizing prompt and continuation, avoiding tokenizer-boundary changes.

Our experiments reveal a more heterogeneous picture than a single “late emergence” statistic suggests. On TruthfulQA, final top-10 visibility is similar across six models (64.7–71.7%), yet the paths differ sharply. Pythia-6.9B first exposes observed targets at 45.4% depth on average but exhibits 28.4% dropout; Gemma-7B first exposes them at 84.9% depth and exhibits 48.0% dropout, with persistent entry at 100.0% depth. Qwen2.5-7B and 14B show similar normalized profiles, but this within-family comparison does not identify a scaling law.

Task-format controls are equally consequential. Qwen2.5-7B exposes every correct MMLU option label in its final top-10, compared with 64.7% of TruthfulQA answer tokens. Across five models on SST-2, candidate accuracy and free-vocabulary visibility can diverge almost completely. These results narrow the interpretation of vocabulary projections and provide concrete reporting require-

ments for future layerwise analyses.

Our contributions are:

- a censoring- and trajectory-aware definition of layerwise target readout, with explicit token-boundary and final-logit consistency checks;
- a complete rerun on six models and three task settings, releasing per-sample ranks rather than only aggregate means;
- evidence that first visibility, persistence, and task competence are distinct, together with a minimal reporting checklist intended to prevent over-interpretation.

2 Related Work

Vocabulary-space projections. The logit lens applies the output unembedding to intermediate states (nostalgebraist, 2020). Geva et al. (2022) study how feed-forward updates promote vocabulary concepts, and Belrose et al. (2023) learn layer-specific affine translators to reduce distribution shift between intermediate and final representations. The tuned lens is more predictive than the raw logit lens, but both remain vocabulary-space probes rather than localizers of stored knowledge. Recent analyses also emphasize that projection behavior can be non-monotone or depend on spectral components poorly represented by a simple token ranking (Cancedda, 2024; Nguyen, 2026). Our focus is complementary: we audit the event statistic computed from such projections.

Factual recall and causal localization. Factual recall involves distributed stages and causally important intermediate modules (Meng et al., 2022; Geva et al., 2023). Model editing work identifies parameters that can alter associations (Meng et al., 2023). A token becoming top- k under an output projection neither establishes where its association is stored nor when it first affects computation. We use “readout” or “visibility” throughout for this reason.

Decoding and early exit. DoLa contrasts layer distributions to improve factual decoding (Chuang et al., 2024); early-exit methods train or detect usable intermediate predictions (Elhoushi et al., 2024). These methods care about intermediate predictions but have different operational objectives. Our metrics diagnose a fixed model under a fixed

prompt and target; they do not themselves provide an exit rule or intervention.

3 Measurement Protocol

3.1 Prompt-conditioned target alignment

For prompt string p_i and gold continuation a_i , we insert exactly one leading space before the stripped continuation and tokenize $p_i + a_i$ jointly. We require that tokenizing p_i alone produces an exact prefix of this sequence. The first continuation token is the target y_i . This matters because tokenizing a_i in isolation can produce a different first token at the boundary.

Let $h_{i,l}$ be the residual state at the final prompt position after transformer block l , for $l \in \{1, \dots, L\}$. We apply the model’s native final normalization and unembedding:

$$z_{i,l} = W_U \text{Norm}_{\text{final}}(h_{i,l}) + b_U, \quad (1)$$

and let $r_{i,l}$ be the zero-based vocabulary rank of y_i . At $l = L$, we verify that the reprojected logits preserve native final top- k membership and differ by at most 0.25 in maximum absolute logit value. Failed samples are retained in an explicit failure log, not silently dropped; all canonical runs reported here have zero failures.

3.2 Trajectory-aware events

For threshold k , define $b_{i,l} = \mathbb{1}[r_{i,l} < k]$. We report:

$$F_i(k) = \min\{l : b_{i,l} = 1\}, \quad (2)$$

$$P_i(k) = \min\{l : \forall m \geq l, b_{i,m} = 1\}. \quad (3)$$

F_i is undefined when the target never enters top- k ; P_i is defined only when it is present at the final layer. We additionally report $N_i = \mathbb{1}[\sum_l b_{i,l} = 0]$ and dropout D_i , which is one when an observed target later leaves top- k . Means of F_i/L are conditional on observed entries; means of P_i/L are conditional on final presence. We always report their denominators alongside unconditional never-observed, final-presence, and dropout rates.

This decomposition prevents three different samples—never observed, final-layer first entry, and early-but-transient entry—from receiving the same scalar value.

4 Experimental Setup

Models. We evaluate Qwen2.5-7B and 14B (Qwen Team, 2024), Llama-3.1-8B (Dubey et al.,

2024), Mistral-7B-v0.1 (Jiang et al., 2023), Pythia-6.9B (Biderman et al., 2023), and Gemma-7B (Gemma Team, Google DeepMind, 2024). All are base models. Model family, tokenizer, training data, architecture, and parameter count are confounded; comparisons are descriptive.

Tasks and targets. TruthfulQA uses all 817 questions and the first token of the reference best-answer continuation (Lin et al., 2022). For MMLU, we sample 1,200 examples balanced across 24 subjects and target the correct option label among A/B/C/D (Hendrycks et al., 2021). For SST-2, we use all 872 validation examples and target the label token negative or positive (Socher et al., 2013). Candidate accuracy is computed by comparing only the task’s permitted label continuations; it is deliberately separated from rank over the full vocabulary.

Implementation. Unless specified otherwise, $k = 10$. Inference uses bfloat16 on one NVIDIA H20 with Python 3.12.13, PyTorch 2.10.0, Transformers 4.48.3, and CUDA 12.8. Artifacts contain prompts, aligned token IDs, ranks and probabilities at every layer, native-logit audit fields, failure records, and SHA-256 checksums. Pure alignment, censoring, persistence, and aggregation functions have CPU unit tests.

5 TruthfulQA Results

Table 1 replaces a single “late” rate with distinct events. Final visibility varies by only seven points across models, but first-entry depth varies from .454 to .872. Figure 1 explains this discrepancy.

Early does not mean stable. Pythia has the earliest conditional first entry (14.53/32 layers), yet 28.4% of all targets drop out after entry and persistent entry moves to 19.67/32. Gemma is more extreme in the opposite direction: observed targets first enter at 23.77/28, 48.0% of all targets drop out at least once, and final-presence cases become persistent at 27.99/28 on average. A scalar first entry calls both targets “observed” while missing their different dynamics.

Never observed is not final-layer entry. Between 21.9% and 31.1% of targets are never top-10 under the native vocabulary projection. Treating these as layer L would inflate conditional depths and create a spurious mass of final-layer events. Conversely, targets genuinely entering at

L remain observed events. Our artifacts keep the two states separate as null and L .

Within-family scale comparison. Qwen2.5-7B and 14B have similar ever-observed rates (68.9% vs. 71.2%) and persistent depths (.876 vs. .900). With only two sizes, this is evidence of profile similarity, not scale invariance.

5.1 Sensitivity to the vocabulary threshold

Increasing k smoothly reduces censoring and shifts entry earlier (Table 2); there is no threshold-independent “emergence layer.” Even at $k = 100$, however, observed targets first enter at 71.0% depth and stabilize at 80.6% depth on average. The appropriate conclusion is therefore conditional: direct vocabulary readout tends to improve late for this prompt and target, while its apparent timing is threshold-dependent.

6 The Interface Is Part of the Measurement

6.1 Controlled prompt-template variation

The abbreviated and expanded neutral templates give similar results (paired exact McNemar $p = .54$ for final visibility), but adding “Answer the following question truthfully” increases final top-10 visibility by 9.7 points (110 gains vs. 31 losses, $p < 10^{-10}$) and shifts persistent entry earlier (Table 3). The model, questions, targets, and threshold are held fixed. Vocabulary readout is therefore not a model-only profile; semantically relevant prompt wording can materially reshape it.

6.2 MMLU option labels

Qwen2.5-7B reaches 72.0% candidate accuracy on 1,200 MMLU examples. Every correct A/B/C/D label both enters and ends in the full-vocabulary top-10. Conditional first and persistent depths are .646 and .731, respectively. Across STEM, humanities, social sciences, and other subjects, final visibility remains 100% while candidate accuracy ranges from 63.1% to 80.0%. Thus, visibility of the correct label is not correctness: several labels can be highly ranked, and task prediction depends on their relative scores.

6.3 SST-2: candidate competence without free readout

Table 4 provides the clearest interpretive stress test. Pythia correctly chooses between negative and

Model	L	Ever $\uparrow$	Never $\downarrow$	Final $\uparrow$	Dropout $\downarrow$	First / Persistent depth
Qwen2.5-7B	28	68.9	31.1	64.7	18.8	.824 / .876
Qwen2.5-14B	48	71.2	28.8	67.7	16.8	.872 / .900
Llama-3.1-8B	32	73.8	26.2	71.7	15.2	.833 / .865
Mistral-7B	32	73.2	26.8	67.9	18.5	.788 / .835
Pythia-6.9B	32	78.1	21.9	71.1	28.4	.454 / .615
Gemma-7B	28	76.5	23.5	71.7	48.0	.849 / 1.000

Table 1: Corrected top-10 trajectories on 817 TruthfulQA examples. Percentages use all examples. First depth is conditional on ever-observed targets; persistent depth is conditional on final top-10 presence. Wilson 95% intervals for the Never rates are at most ± 3.3 points and are given in the released table.

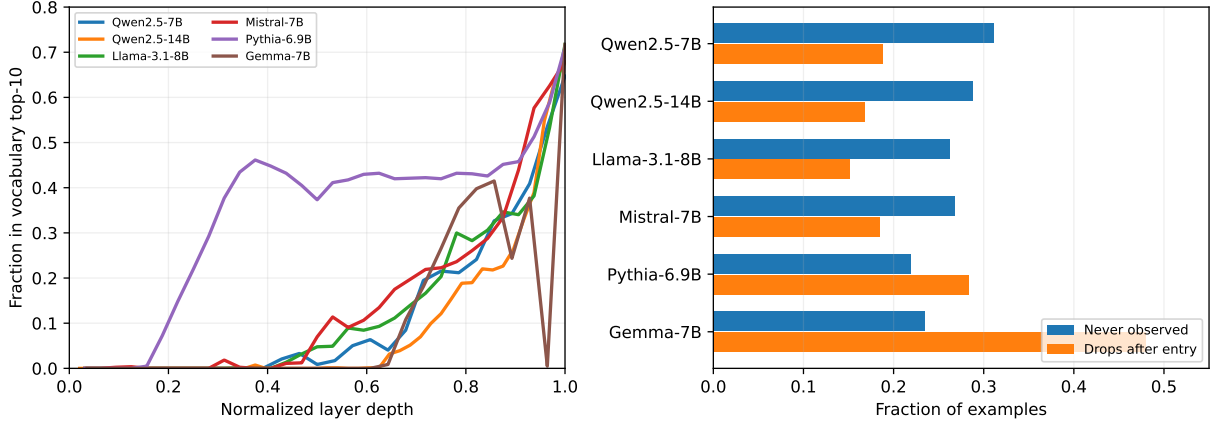


Figure 1: TruthfulQA top-10 trajectories. Left: fraction of target tokens in the vocabulary top-10 at each normalized depth. Right: right-censoring and non-monotonicity are both substantial. Pythia’s early visibility plateau and Gemma’s late collapse show why first entry alone is insufficient.

k	Ever	Never	Final	First / Persist.
1	27.3	72.7	19.7	.880 / .941
3	49.2	50.8	40.0	.850 / .905
5	57.5	42.5	50.9	.836 / .890
10	68.9	31.1	64.7	.824 / .876
20	76.9	23.1	74.7	.800 / .859
50	85.9	14.1	84.7	.758 / .831
100	91.8	8.2	90.2	.710 / .806

Table 2: Threshold sensitivity for Qwen2.5-7B on TruthfulQA (percent except normalized depths). Conclusions depend quantitatively on k , so a single threshold should not be presented as intrinsic.

Template	Ever	Final	Dropout	First / Persist.
Q/A	68.9	64.7	18.8	.824 / .876
Question/Answer	69.6	65.7	22.3	.818 / .865
Truthful instruction	77.2	74.4	16.6	.811 / .842

Table 3: Qwen2.5-7B on the same 817 TruthfulQA questions and targets under three prompt templates (percent except normalized depths).

positive on 80.8% of examples, but neither gold-label behavior nor competence is reflected by FEP: the gold label is never in the full-vocabulary top-10. Mistral obtains 86.9% candidate accuracy; its gold labels enter top-10 for 48.9% of examples and then disappear in every such case. These are not contradictions. Candidate accuracy asks which permitted label scores higher, whereas FEP asks whether one label outranks almost the entire vocabulary under a free-form prompt.

The same target strings also produce sharply different trajectories across models. Consequently,

comparing FEP across architectures without controlling tokenization, prompt calibration, and answer space can attribute interface differences to internal computation.

7 Implications for Mechanistic Analysis

What the measure supports. For a fixed model, prompt, target construction, projection, and k , the trajectory describes when a token is directly visible through the output head. It can identify early stable readout, transient competition, late commitment, and non-entry. These are useful observational phenotypes.

What it does not support. The trajectory does not show that information is absent before entry, stored at the entry layer, or first used causally there.

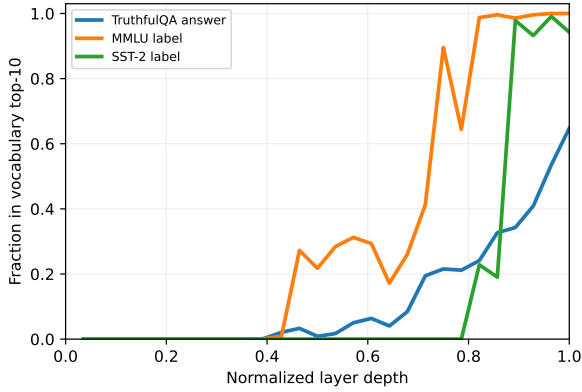


Figure 2: Qwen2.5-7B top-10 prevalence for three targets. MMLU option labels become visible much earlier than TruthfulQA answer tokens; SST-2 labels change sharply near the output.

Model	Acc.	Ever	Final	Dropout
Qwen2.5-7B	91.6	99.5	94.3	20.5
Mistral-7B	86.9	48.9	0.0	48.9
Llama-3.1-8B	73.2	44.5	35.1	28.4
Pythia-6.9B	80.8	0.0	0.0	0.0
Gemma-7B	69.8	2.1	2.1	0.0

Table 4: SST-2 candidate accuracy versus full-vocabulary top-10 visibility (872 examples; percentages). A model can choose correctly between two labels even when the gold label is never a likely unconstrained continuation.

Linear or nonlinear probes may recover task information that the native unembedding does not expose, and causal interventions answer a different question. Candidate-task performance may also be high despite non-entry (Table 4). We therefore avoid the language of knowledge “appearing” or “crystallizing” without the qualifier *vocabulary-space readout*.

Consequences for intervention claims. The corrected TruthfulQA profiles do not preserve the architecture ordering previously suggested by a scalar final-layer statistic. Qwen2.5-7B, Llama-3.1-8B, and Mistral-7B have close first-entry depths (.824, .833, and .788) despite different intervention outcomes in preliminary experiments. FEP should therefore not be used as an intervention-selection rule without a prospective, controlled study. We exclude FEP-guided intervention, LayerNorm-boundary, and head-at-boundary results from the present claims because their layer choices depended on the audited scalar.

Recommended reporting checklist. Layerwise vocabulary-readout studies should: (1) tokenize

the full prompt and continuation jointly; (2) distinguish never observed from final-layer entry; (3) report persistence or the complete trajectory, not only first entry; (4) validate the last projected state against native logits; (5) report sensitivity to k ; (6) separate candidate-set scores from full-vocabulary ranks; and (7) name the projection explicitly and avoid causal or storage language unless independently tested.

8 Conclusion

Layerwise vocabulary projections are informative, but their simplest event summary can be misleading. Across six language models, censoring and non-monotonicity are common, and models with similar final visibility follow qualitatively different paths. Across task formats, candidate competence can be nearly independent of full-vocabulary top- k visibility. A trajectory-aware, censoring-aware protocol turns these apparent contradictions into measurable distinctions. The broader lesson is methodological: vocabulary readout should be treated as an interface-sensitive observation, not as a timestamp for knowledge itself.

Limitations

We analyze one raw logit-lens projection, one prompt template per task, English benchmarks, base models, and only the first continuation token. The results do not establish how trained probes, alternative prompts, instruction tuning, multilingual inputs, or multi-token generation change the trajectories. Conditional means describe only observed or final-present subsets and should not be compared without their censoring rates. Our model comparisons are observational because architecture, data, optimization, tokenizer, and scale vary jointly. The MMLU sample is balanced over 24 subjects but is not the full benchmark. Finally, rank thresholds discard score margins; future work should jointly model rank, probability, and candidate-relative evidence.

Ethics Statement

This work audits an interpretability measurement and uses public benchmarks and openly released model weights. It makes no claims about individual people and introduces no new personal data. More reliable interpretation can reduce misleading claims about model internals, but vocabulary projections may still be over-trusted; we state their

limits explicitly.

Reproducibility Statement

The supplementary artifact includes the exact runner, pure measurement helpers, unit tests, compressed per-sample JSON, aggregate CSV tables, figure-generation code, environment versions, and SHA-256 checksums. All random sampling uses seed 42. Canonical files record model references, dtype, prompt, aligned continuation IDs, target IDs, ranks and probabilities for every layer, candidate predictions where applicable, final-logit audit fields, and failures.

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